Lab 6 Report

Minimal solvers, RANSAC and frame-to-frame motion estimation

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Abstract

Frame-to-frame motion estimation must recover the camera pose from feature matches that contain outliers. We derive the minimal solvers behind this task, Nistér's five-point algorithm and a two-point solver for known rotation, together with the RANSAC iteration counts they imply. We then evaluate four solvers inside RANSAC on all 3 887 consecutive frame pairs of a simulated office flight. A sub-pixel inlier threshold of 0.1 px brings the five-point solver to a median rotation error of 0.034°, ten times lower than at 1 px. Prior information gives the largest further gain. A known rotation lowers the median translation-direction error from 0.066 to 0.020 at 0.39 ms per frame, and depth lets Arun's 3D–3D method reach 0.019° and 1.7 mm. Without RANSAC, the same matches give rotation errors of 0.6–14°.

1 Individual work

Notation. Two calibrated views are taken at times t_1 and t_2. A correspondence is a pair of unit bearing vectors (f_1, f_2) that point to the same 3D point from the two camera centres. The relative pose (R, t) maps coordinates in frame 2 to frame 1, $p_1 = Rp_2 + t$: R rotates frame 2 into frame 1, and t is the centre of camera 2 expressed in frame 1. Since $\lambda_1 f_1 = \lambda_2 R f_2 + t$, the vectors f_1, Rf_2 and t are coplanar, which gives the epipolar constraint

$$f_1^\top E f_2 = 0, \qquad E = [t]_\times R. \tag{1}$$

OpenGV [1], used in the team part, follows the same convention.

1.1 Deliverable 1: Nistér's five-point algorithm

1. Computational steps. Nistér's solver [2] takes five correspondences and returns every essential matrix consistent with them, at most ten, each with its pose (R, t) up to scale.

1. *Null space of the linear constraints.* With E stored row by row as $\tilde{E} \in \mathbb{R}^9$, each correspondence turns (1) into $(f_1 \otimes f_2)^\top \tilde{E} = 0$. Five correspondences give a 5×9 matrix with a four-dimensional null space. Its basis X, Y, Z, W (from QR or SVD) parametrises every candidate as $E = xX + yY + zZ + W$.

2. *Cubic constraints.* A valid essential matrix has two equal singular values and one zero singular value, which is equivalent to

$$\det E = 0, \qquad 2EE^\top E - \operatorname{tr}(EE^\top)E = 0. \tag{2}$$

   Substituting the parametrisation gives ten cubic equations in (x, y, z), written as a 10×20 matrix over the 20 monomials of degree at most 3.

3. *Gauss–Jordan elimination.* Eliminating this matrix with a suitable monomial order leaves rows that Nistér combines into three polynomials $\langle k \rangle$, $\langle l \rangle$ and $\langle m \rangle$. Together they form a 3×3 matrix $B(z)$, polynomial in z, with $B(z)\,[x\ y\ 1]^\top = 0$.

4. *Tenth-degree polynomial.* A non-trivial solution requires $\det B(z) = 0$, a univariate polynomial of degree 10. Its real roots are bracketed with a Sturm sequence and refined by bisection or Newton steps.

5. *Back-substitution.* For each real root z, the null vector of $B(z)$ gives (x, y), and hence $E = xX + yY + zZ + W$.

6. *Pose recovery.* The SVD $E = U\,\mathrm{diag}(1, 1, 0)V^\top$ gives $t \propto \pm u_3$ and $R \in \{UDV^\top, UD^\top V^\top\}$ with $D = \left[\begin{smallmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right]$. Of these four poses, the cheirality test keeps the one that places a triangulated point in front of both cameras.

7. *Selection.* The remaining hypotheses, one per real root, are scored on further correspondences, and RANSAC keeps the pose with the largest consensus set.

2. Degeneracies. The five-point solver handles a planar scene, which defeats the linear methods. With coplanar points the $N \times 9$ system of the eight-point algorithm loses rank. The five-point solver enforces (2) and still returns a finite solution set containing the true pose [2]. Three configurations remain degenerate for any two-view method:

- *Zero baseline (pure rotation).* With $t = 0$ every correspondence satisfies $f_1 \parallel Rf_2$, and $E = 0$. The translation direction is then unobservable, and R cannot be recovered from E. Points much farther away than the baseline approach this case numerically.

- *Critical surfaces.* When the 3D points and both camera centres lie on a ruled quadric, such as a hyperboloid of one sheet or a pair of planes, several essential matrices explain the data exactly. A plane gives a two-fold ambiguity. Correspondences off the surface resolve it, which is what RANSAC scoring provides.

- *Dependent correspondences.* Collinear points or repeated correspondences make the five equations linearly dependent, so the null space has more than four dimensions.

3. Expected number of RANSAC iterations. Following the lecture notes, let ϵ be the outlier ratio and s the size of the minimal set, and assume the samples of different iterations are independent. A sample is outlier-free with probability $w = (1 - \epsilon)^s$. The number of iterations until the first outlier-free sample is therefore geometric, with expectation

$$\mathbb{E}[k] = \frac{1}{(1 - \epsilon)^s}. \tag{3}$$

To draw at least one outlier-free sample with probability p, RANSAC needs

$$N = \left\lceil \frac{\log(1 - p)}{\log\bigl(1 - (1 - \epsilon)^s\bigr)} \right\rceil \tag{4}$$

iterations. For the five-point solver ($s = 5$) at $\epsilon = 50\,\%$, $\mathbb{E}[k] = 2^5 = 32$ and $N = 146$ for $p = 0.99$. OpenGV draws three extra correspondences per sample to choose among the up to ten solutions, so in practice $s = 8$ and N grows to 1 177. Table 1 compares the solvers used in this lab.

Table 1: RANSAC iterations versus minimal-set size s and outlier ratio ϵ: expected iterations to the first outlier-free sample, $1/(1-\epsilon)^s$, and the number N in (4) for $p = 0.99$.

		$\epsilon = 10\,\%$		$\epsilon = 30\,\%$		$\epsilon = 50\,\%$		$\epsilon = 70\,\%$	
Solver	s	$\mathbb{E}[k]$	N	$\mathbb{E}[k]$	N	$\mathbb{E}[k]$	N	$\mathbb{E}[k]$	N
2-point (known R)	2	1.2	3	2.0	7	4	17	11.1	49
3-point Arun (3D–3D)	3	1.4	4	2.9	11	8	35	37.0	169
5-point Nistér	5	1.7	6	5.9	26	32	146	411.5	1 893
8-point Longuet-Higgins	8	2.3	9	17.3	78	256	1 177	15 241.6	70 188

1.2 Deliverable 2: A two-point minimal solver with known rotation

Minimal solver. With R known from the gyroscope, only the translation remains. Images observe only its direction, $t \in \mathbb{S}^2$, which has two degrees of freedom. Writing (1) as a scalar triple product,

$$f_1^\top [t]_\times R f_2 = f_1 \cdot (t \times R f_2) = t \cdot (R f_2 \times f_1) = 0, \tag{5}$$

shows that each correspondence gives one linear homogeneous equation $a_i^\top t = 0$. The known vector $a_i = R f_{2,i} \times f_{1,i}$ is the normal of the epipolar plane of point i. Two correspondences are therefore minimal, and t is orthogonal to both normals:

$$t = \pm \frac{a_1 \times a_2}{\|a_1 \times a_2\|}. \tag{6}$$

Cheirality fixes the sign. Solving $\lambda_1 f_1 - \lambda_2 R f_2 = t$ for one point gives its two depths, and the sign with $\lambda_1 > 0$ and $\lambda_2 > 0$ is kept. The solver costs two cross products and one 2×2 solve, and it returns a single solution instead of up to ten. It fails when both points lie in the same epipolar plane ($a_1 \parallel a_2$) and when $t = 0$, where every a_i vanishes. With $N > 2$ correspondences, the least-squares estimate of t is the right singular vector of $A = [a_1 \cdots a_N]^\top$ with the smallest singular value.

RANSAC with the two-point solver (optional). Algorithm 1 uses (6) to generate hypotheses. The residual $r_i = |a_i^\top t| / \|a_i\|$ is the sine of the angle between t and the epipolar plane of point i. OpenGV scores the reprojection error of the triangulated point instead, which works equally well. Because $s = 2$, RANSAC needs 17 iterations at $\epsilon = 50\,\%$ and $p = 0.99$, against 146 for the five-point solver (Table 1). The final least-squares step on the consensus set is the refit we use in the team part (Section 2.1).

Algorithm 1 RANSAC for the translation direction with known rotation

Require: correspondences $\{(f_{1,i}, f_{2,i})\}_{i=1}^{N}$, rotation R, threshold τ, confidence p, iteration cap $K_{\max}$
$a_i \leftarrow R f_{2,i} \times f_{1,i}$ for all i; $\mathcal{I}^\star \leftarrow \emptyset$; $K \leftarrow K_{\max}$; $k \leftarrow 0$
while $k < K$ **do**
  draw two distinct indices i, j uniformly at random
  if $\|a_i \times a_j\|$ is below a small tolerance **then**
    $k \leftarrow k + 1$; **continue** ▷ degenerate sample
  end if
  $t \leftarrow a_i \times a_j / \|a_i \times a_j\|$; flip the sign of t if point i fails the cheirality test
  $\mathcal{I} \leftarrow \{\, m : |a_m^\top t| / \|a_m\| < \tau$ and point m passes the cheirality test $\}$
  if $|\mathcal{I}| > |\mathcal{I}^\star|$ **then**
    $\mathcal{I}^\star \leftarrow \mathcal{I}$; $w \leftarrow |\mathcal{I}|/N$
    $K \leftarrow \min\big(K_{\max}, \lceil \log(1-p) / \log(1-w^2) \rceil\big)$ ▷ adaptive stop
  end if
  $k \leftarrow k + 1$
end while
$t^\star \leftarrow$ smallest right singular vector of $[a_m^\top]_{m \in \mathcal{I}^\star}$, sign fixed by cheirality on $\mathcal{I}^\star$
return $t^\star$, $\mathcal{I}^\star$

2 Team work

2.1 Pipeline and evaluation (Deliverable 3)

Data and correspondences. The office sequence contains 3 890 RGB-D frames over 237 s from the simulated drone, with ground-truth odometry at 200 Hz. The camera has $f = 415.7$ px and a 720×480 image, with no distortion. For each consecutive frame pair, the Lab 5 SIFT tracker returns matches that passed Lowe's ratio test, outliers included. `calibrateKeypoints` undistorts them with K and the distortion coefficients and converts them to unit bearing vectors, which feed the OpenGV adapters.

Estimators. All four estimators run inside OpenGV's RANSAC:

- *5-point* (`NISTER`) and *8-point* (`EIGHTPT`) solve for (R, t) with `CentralRelativePoseSacProblem`;
- *2-point* solves for t with `TranslationOnlySacProblem`, given the ground-truth relative rotation;
- *3-point Arun* aligns the two point clouds obtained by scaling the bearing vectors with the registered depth, using `PointCloudSacProblem`.

OpenGV's RANSAC returns the model of the best minimal sample. We refit it on the consensus set: nonlinear optimisation for the 5- and 8-point models, Arun's method on all inliers for the 3D–3D model, and the least-squares translation of Deliverable 2 for the 2-point model, which keeps the given rotation fixed. The optional baselines without RANSAC fit the 5-point, 8-point and Arun models to all matches at once. RANSAC runs at most 5 000 iterations.

Ground truth and relative pose error. The ground-truth camera pose is $T^W_{\text{cam}} = T^W_{\text{body}}\, T^{\text{body}}_{\text{cam}}$, interpolated at each image timestamp (linear in position, spherical in rotation). For each frame pair the estimate $T^{t-1}_{\text{est},t}$ is compared with $T^{t-1}_{\text{ref},t} = (T^W_{\text{ref},t-1})^{-1} T^W_{\text{ref},t}$. The rotation error is the chordal distance $\|R_{\text{ref}} - R_{\text{est}}\|_F$, reported as the equivalent angle $\theta = 2 \arcsin\big(\|R_{\text{ref}} - R_{\text{est}}\|_F / 2\sqrt{2}\big)$. The translation error is $\|t_{\text{ref}} - t_{\text{est}}\|_2$ between unit vectors, which lies in $[0, 2]$. For Arun's method we also report the metric error in millimetres. The node publishes $T^W_{\text{ref},t-1}\, T^{t-1}_{\text{est},t}$ on `/camera_pose`, with the translation rescaled to the ground-truth step length for the 2D–2D methods, next to the ground truth on `/gt_camera_pose` for RViz.

2.2 Two-view estimators with RANSAC (Deliverable 4)

Accuracy over the sequence. All three 2D–2D estimators track the camera motion over the full sequence (Fig. 1, Table 2). The 5-point solver has a median rotation error of 0.034°, and 99.8 % of its 3 887 frame pairs stay below 1°. The camera moves about 3 cm between frames, so the parallax at a depth of a few metres is only a few pixels. In this regime the translation direction is the harder quantity, and the 5-point solver recovers it with a median unit-vector error of 0.066 (about 3.8°). The 8-point solver performs similarly, with a median rotation error of 0.039° and a median translation error of 0.087. The errors of both solvers rise together in a few short episodes. Near 120 s the camera moves only 8 mm per frame, and near 195 s the median number of matches drops from 352 to 246.

The inlier threshold sets the accuracy. The simulator renders sharp images, and SIFT localises keypoints to a fraction of a pixel. A threshold far below the usual 1 px therefore keeps only the most precise matches, and it improves every estimator (Fig. 2). Going from 1 px to 0.1 px lowers

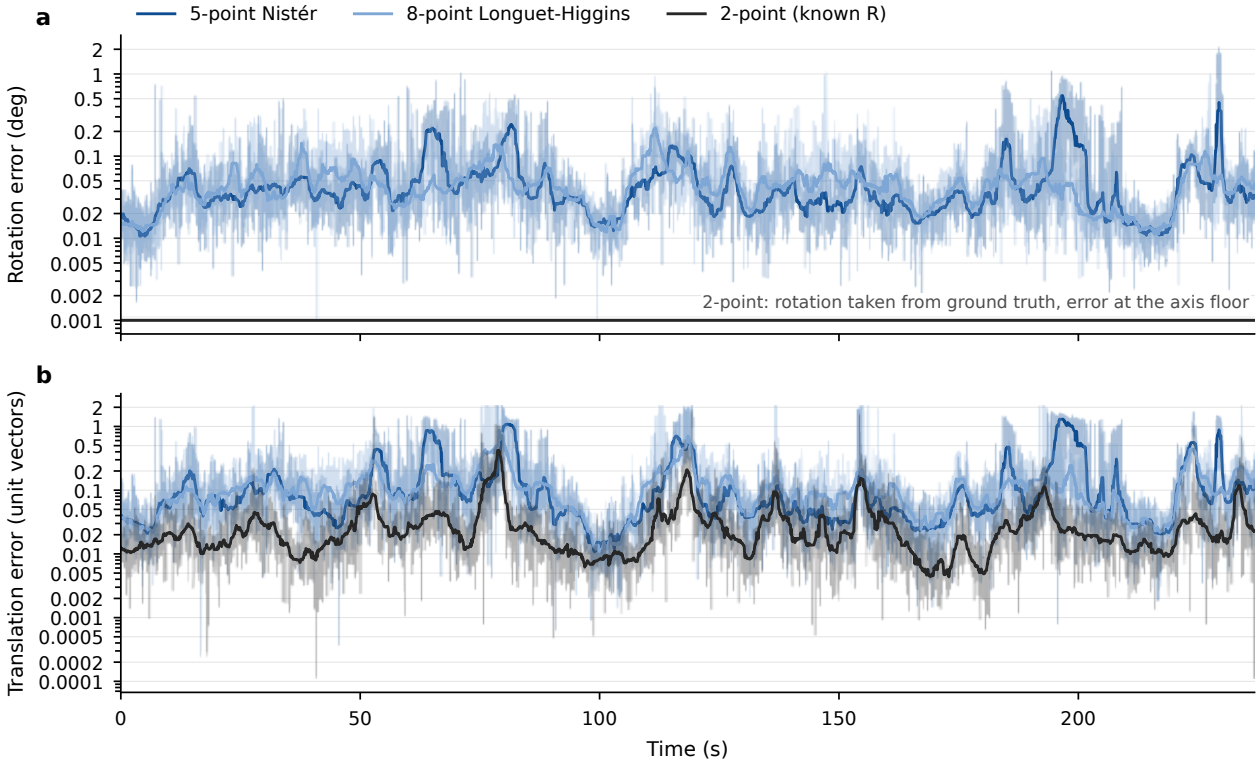


Figure 1: Relative pose error of the 2D–2D estimators over the office sequence. **a**, Rotation error. **b**, Translation-direction error between unit vectors. Thin lines show every frame pair; thick lines show a centred 2 s rolling median. RANSAC threshold 0.1 px, at most 5 000 iterations, $n = 3\,887$ frame pairs per method.

Table 2: Relative pose error over the full sequence ($n = 3\,887$ frame pairs). Rotation is the equivalent angle of the chordal distance; translation is the distance between unit direction vectors. Inlier ratio and solver time are per-frame medians. RANSAC thresholds are 0.1 px for the 2D–2D methods and 1 cm for Arun's method.

	Rotation		Translation direction			RANSAC	
Method	median	$< 1°$	median	< 0.5	< 0.1	inliers	time
5-point Nistér	0.034°	99.8 %	0.066	89.4 %	62.3 %	49 %	85 ms
8-point Longuet-Higgins	0.039°	99.97 %	0.087	95.1 %	55.0 %	45 %	140 ms
2-point (known R)	—	—	0.020	99.4 %	92.9 %	48 %	0.39 ms
3-point Arun (3D–3D)	0.019°	100 %	0.044	98.3 %	82.3 %	41 %	0.25 ms
5-point, no RANSAC	14.0°	0.3 %	1.27	4.3 %	0.1 %	—	6.1 ms
8-point, no RANSAC	1.17°	42.5 %	0.77	39.1 %	4.9 %	—	2.0 ms
Arun, no RANSAC	0.61°	79.5 %	1.05	17.4 %	0.8 %	—	0.03 ms

the median translation error of the 5-point solver from 0.59 to 0.063 and its median rotation error from 0.36° to 0.031°, while the median inlier ratio falls from 86 % to 48 %. The 5-point solver gains the most from the tighter threshold, and at 0.1 px it is the more accurate of the two solvers that estimate rotation and translation together. We use 0.1 px for all results.

Known rotation. Given the rotation, the 2-point solver has a median translation error of 0.020, three times lower than the 5-point solver, and 92.9 % of its frame pairs are below 0.1. Its rotation error is zero up to floating point because the rotation is taken from ground truth; on a real drone it would come from the gyroscope, and the gyroscope error would add to it. The solver is also more than 200 times faster than the 5-point solver (0.39 ms against 85 ms per frame), because each hypothesis costs two cross products and RANSAC needs far fewer samples.

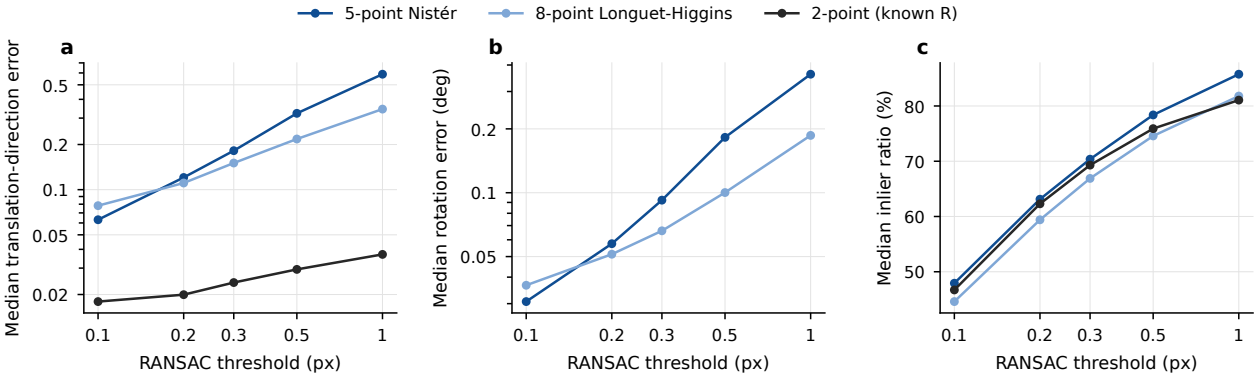


Figure 2: Effect of the RANSAC inlier threshold, measured on the first 60 s of the sequence ($n \approx 1\,040$ frame pairs per point). **a**, Median translation-direction error. **b**, Median rotation error of the 5- and 8-point solvers (the 2-point solver takes the rotation as input). **c**, Median inlier ratio.

Pose publication. For every frame pair the node publishes the estimate on `/camera_pose`, chained onto the previous ground-truth pose (Section 2.1), and the ground truth on `/gt_camera_pose`. RViz shows both for any estimator selected with the `pose_estimator` flag.

2.3 3D–3D registration with Arun's method (Deliverable 5)

With the depth image, each match becomes a pair of 3D points and the relative pose is the rigid transform between two point clouds. Three points determine it, and Arun's SVD solution [3] gives it in closed form. Depth also fixes the metric scale, which no 2D–2D method observes. Arun's method has the lowest rotation error of the estimators that solve for rotation, a median of 0.019°, and every frame pair stays below 1° (Fig. 3, Table 2). The median metric translation error is 1.7 mm for a median step of 31 mm. A threshold sweep on the first 60 s gives the same rotation accuracy for thresholds between 0.5 and 2 cm (0.016–0.017°); at 10 cm the median rotation error rises to 0.031° and the translation error to 3.4 mm. We use 1 cm. With a minimal set of three, RANSAC is cheap, and the whole estimate takes 0.25 ms per frame.

2.4 What RANSAC contributes (optional baselines)

Fig. 4 places all seven estimators on common axes. Fitting a model to every ratio-test match, outliers included, moves the error distribution by one to two orders of magnitude. The 5-point solver without RANSAC fits Nistér's polynomial system to all matches and keeps the solution that places the most points in front of both cameras; its median rotation error reaches 14°. The least-squares 8-point and Arun fits degrade more gracefully, with median rotation errors of 1.2° and 0.6°, but their translation directions are close to random (medians of 0.77 and 1.05 on a scale of 0 to 2). With RANSAC, the same matches give rotation errors below 0.1° for more than 80 % of frame pairs with every solver. The ordering of the RANSAC curves also shows the value of prior information: the 2-point solver, which knows the rotation, and Arun's method, which knows the depth, both lie to the left of the 5- and 8-point solvers in translation.

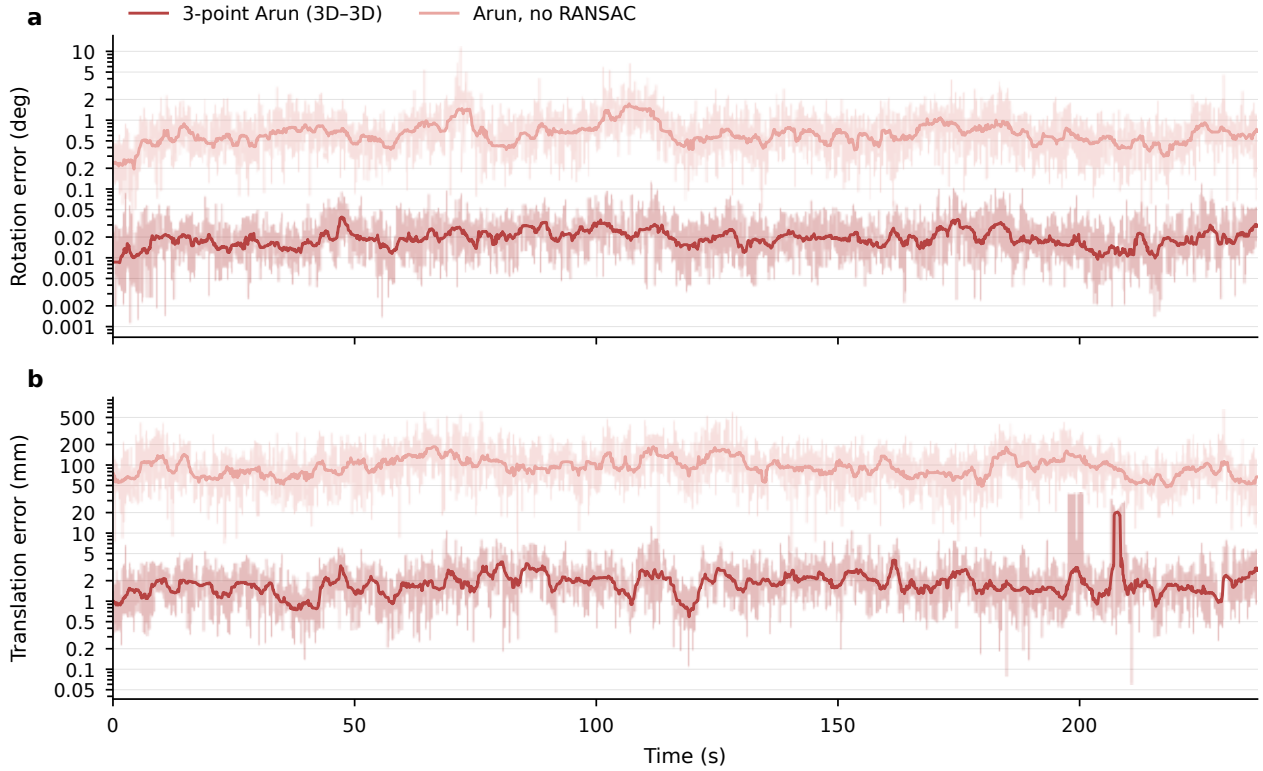


Figure 3: Relative pose error of Arun's 3D–3D method over the office sequence, with RANSAC (1 cm threshold) and fitted to all matches without RANSAC. **a**, Rotation error. **b**, Metric translation error. Thin lines show every frame pair; thick lines show a centred 2 s rolling median; $n = 3\,887$.

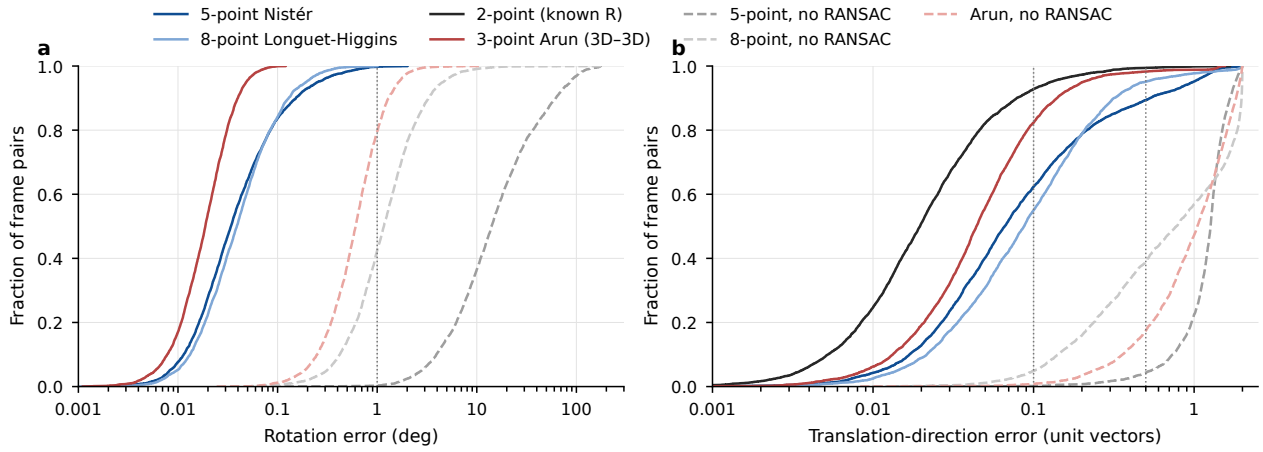


Figure 4: Empirical cumulative distributions of the relative pose error for all estimators over the full sequence ($n = 3\,887$ frame pairs each). Solid lines use RANSAC; dashed lines fit all matches at once. **a**, Rotation error; the 2-point solver is omitted because its rotation is given. **b**, Translation-direction error. Dotted lines mark 1°, 0.1 and 0.5.

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[2] David Nistér. An efficient solution to the five-point relative pose problem. In *IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR)*, volume 2, pages II–195–II–202, 2003.

[3] K. Somani Arun, Thomas S. Huang, and Steven D. Blostein. Least-squares fitting of two 3-D point sets. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 9(5):698–700, 1987.